
**Media Enhancement Coprocessor for the
Super Nintendo Entertainment System**

High-Quality Audio Playback:

- Standard Red Book audio
- Stereo
- 16-bit, signed samples
- 44.1KHz sampling rate
- Gapless looping
- Supports single-track playback resume^[1]

High-Capacity Storage Expansion:

- Up to 4Gbyte addressable storage
- Supports fixed-source DMA
- 3.58Mbyte/s transfer rate

Simple Access Interface:

- 8 read-only and 8 write-only registers

^[1] Playback resume feature only available on revision 2 or later

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1.0 Device Overview

Text

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2.0 Audio

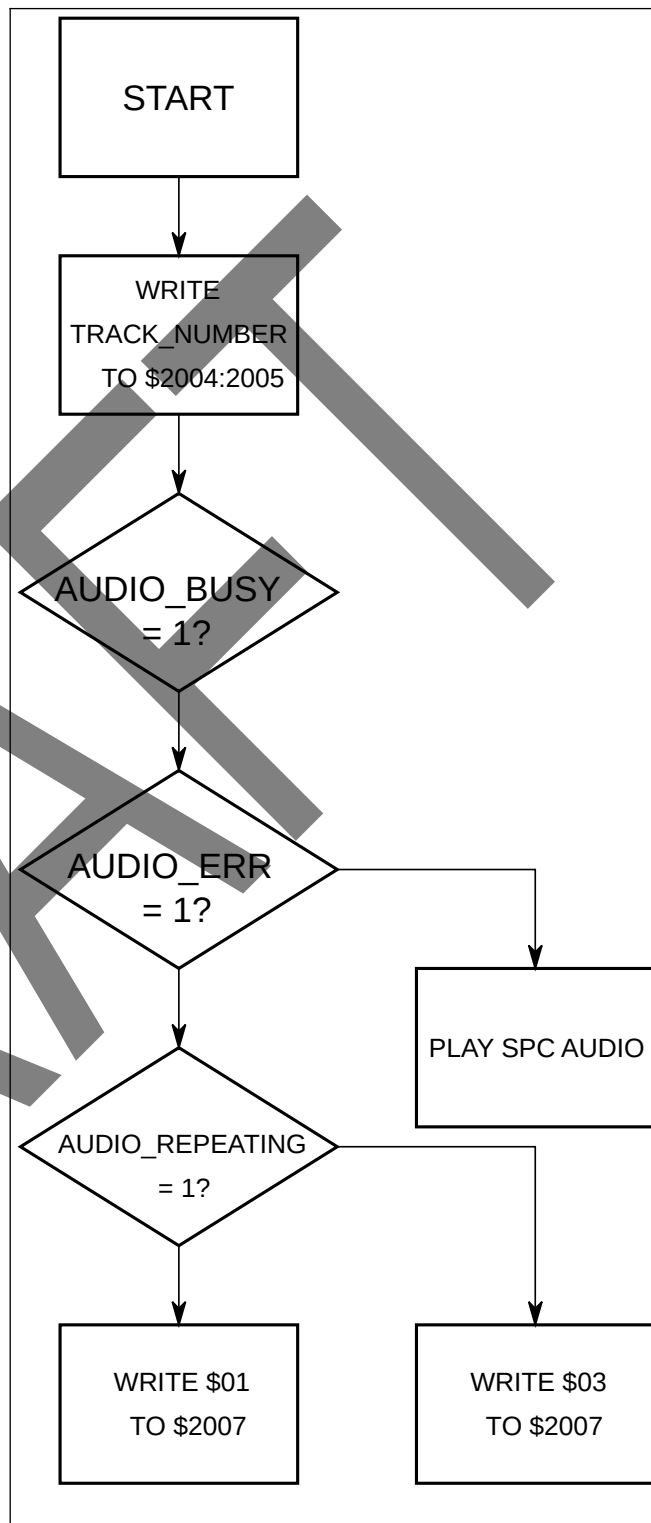
The audio playback interface is comprised of three write-only registers. These registers are:

- MSU_TRACK
- MSU_VOLUME
- MSU_CONTROL

It also includes four flags contained in the read-only MSU_STATUS register. These flags are:

- AUDIO_BUSY
- AUDIO_PLAYING
- AUDIO_REPEATING
- AUDIO_ERROR

FIGURE 2-1: TRACK LOAD FLOWCHART



3.0 Data

The data streaming interface is comprised of a single write-only register and a single read-only register. These registers are:

- MSU_SEEK
- MSU_DATA

It also includes one flag contained in the read-only MSU_STATUS register.

- DATA_BUSY

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4.0 Register Specification

4.1 Read-Only Registers

4.1.1 MSU_STATUS (\$2000)

R-1	R-0	R-0	R-0	R-0	R-0	R-1	R-0
DATA_BUSY	AUDIO_BUSY	REPEATING	PLAYING	AUDIO_ERR		REVISION<2:0>	
bit 7							bit 0

- bit 7 **DATA_BUSY:** Data load busy flag
1 = Data buffer is not full
0 = Data buffer full, ready to read
- bit 6 **AUDIO_BUSY:** Audio track loading busy flag
1 = Audio track loading
0 = Audio track load complete, or failed (see bit 4)
- bit 5 **REPEATING:** Audio repeating
1 = Audio track is repeating
0 = Audio track is not repeating
- bit 4 **PLAYING:** Audio playing
1 = Audio track is playing
0 = Audio track is not playing
- bit 3 **AUDIO_ERR:** Audio track loading error
1 = Audio track not found
0 = No error
- bit 2-0 **REVISION<2:0>:** Chip revision
See Appendix A

MSU-1 (rev. 2)

4.1.2 MSU_DATA (\$2001)

R-0	R-0	R-0	R-0	R-0	R-0	R-0	R-0
DATA<7:0>							
bit 7 bit 0							

bit 7-0 **DATA:** MSU-1 data register^[1]

4.1.3 MSU_ID (\$2002-\$2007)

R-1	R-0	R-1	R-0	R-0	R-0	R-1	R-1
'S'							
R-0	R-0	R-1	R-0	R-1	R-1	R-0	R-1
'U'							
R-0	R-1	R-0	R-0	R-1	R-1	R-0	R-1
'M'							
R-1	R-0	R-1	R-0	R-0	R-0	R-1	R-1
'S'							
R-0	R-1	R-0	R-1	R-0	R-1	R-0	R-1
'U'							
R-0	R-0	R-1	R-1	R-0	R-0	R-0	R-1
'1'							
bit 7				bit 0			

bit 7-0: **ID:** Static chip identification bytes. ASCII "S-MSU1"

Note 1: Reading from MSU_DATA automatically advances the read pointer by 1

4.2 Write-Only Registers

4.2.1 MSU_SEEK (\$2000-\$2003)

W-0	W-0	W-0	W-0	W-0	W-0	W-0	W-0
ADDR[0]<7:0>							
ADDR[1]<7:0>							
ADDR[2]<7:0>							
ADDR[3]<7:0>							
bit 7				bit 0			

bit 7-0 **ADDR<7:0>**: 32-bit address offset, little-endian^[1]

4.2.2 MSU_TRACK (\$2004-\$2005)

W-0	W-0	W-0	W-0	W-0	W-0	W-0	W-0
TRACK[0]							
TRACK[1]							
bit 7				bit 0			

bit 7-0 **TRACK<7:0>**: 16-bit track number, little-endian^[2]

4.2.3 MSU_VOLUME (\$2006)

W-0	W-0	W-0	W-0	W-0	W-0	W-0	W-0
VOLUME<7:0>							
bit 7				bit 0			

bit 7-0 **VOLUME<7:0>**: Audio volume level

4.2.4 MSU_CONTROL (\$2007)

U-0	U-0	U-0	U-0	U-0	R-0	R-0	R-0
-	-	-	-	-	RESUME	REPEAT	PLAY
bit 7					bit 0		

bit 7-3 **Unimplemented:** Reserved for future use

bit 2 **RESUME:** Update playback resume data

1 = Store the playback position for the currently loaded track

Note: Has no effect if bit 1 or 0 is also set at the same time

0 = Playback position is not stored

bit 1 **REPEAT:** Enable gapless looping

1 = Repeat is enabled

0 = Repeat is disabled

bit 0 **PLAY:** Enable playback

1 = Play currently loaded song

0 = Pause playback

Note 1: Writing to \$2003 latches the full address and initiates the seek operation
Note 2: Writing to \$2005 latches the full track number and initiates the track load operation

Appendix A: Chip Revisions

Revision 1

- Initial release
- Revision field (bits 2-0) of the MSU_STATUS register set to 001

Revision 2

- Audio resume feature added
- Audio resume flag (bit 2) added to the MSU_CONTROL register
- Revision field (bits 2-0) of the MSU_STATUS register updated to 010

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Appendix B: PCM Audio File Specification

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Appendix C: End User Considerations

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Appendix D: Errata

SD2SNES

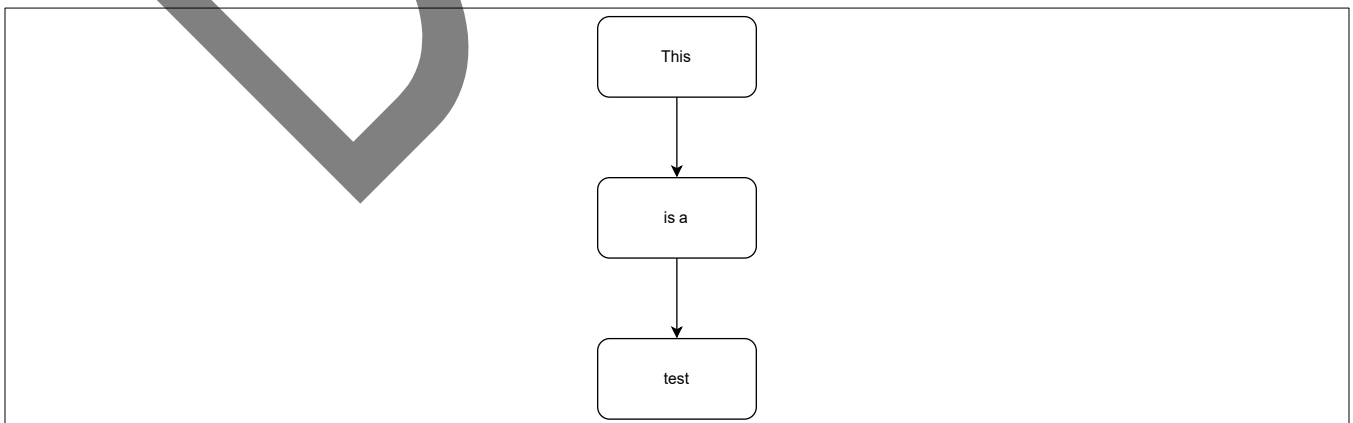
Rapid writes to MSU_CONTROL and MSU_TRACK may not be processed in the order they were received. After writing \$04 to MSU_CONTROL to set the resume flag, ensure that the PLAYING flag in MSU_STATUS is clear before trying to write a new track number to MSU_TRACK.

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Column 1:

Column 2:

FIGURE:



Note 1:
2:
3:

; Code
; Block

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